IPCON® ASA CF

Carbon Fiber Reinforced Acrylonitrile-Styrene-Acrylate Copolymer Filament

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.05 mm
Net Filament Weight	0.75 kg ± 0.5%
Filament Length	300 m ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C)
Spool Size	Outer Cardboard's diameter: 200 mm; Inner roller's diameter: 90 mm

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.04 g/cm³
Shrinkage Rate*	Specially-defined	0.14% - 0.32%
Surface Hardness	Shore hardness	81D
Saturated Water Absorption Rate	25 °C, 55% RH, room air	0.33%
Melt Index	290 °C, 2.16 kg	17.7 g/10 min
Melting Temperature	DSC, 10 °C/min	223 °C
Vicar Softening Temperature	ISO 306, GB/T 1633	109 °C
Heat Deflection Temperature	ISO 75, 0.45 MPa	99 °C

Mechanical Properties*	Testing Methods	Data*
Tensile Strength XY	ISO 527, GB/T 1040	44 MPa
Tensile Strength Z	ISO 527, GB/T 1040	27 MPa
Young's Modulus XY	ISO 527, GB/T 1040	3352 MPa
Young's Modulus Z	ISO 527, GB/T 1040	2373 MPa
Breaking Elongation Rate XY	ISO 527, GB/T 1040	6.1%
Breaking Elongation Rate Z	ISO 527, GB/T 1040	2.3%
Flexural Strength XY	ISO 178, GB/T 9341	80 MPa
Flexural Strength Z	ISO 178, GB/T 9341	45 MPa
Flexural Modulus XY	ISO 178, GB/T 9341	3171 MPa
Flexural Modulus Z	ISO 178, GB/T 9341	2173 MPa
Impact Strength XY	ISO 179, GB/T 1043	25.7 kJ/m²
Impact Strength Z	ISO 179, GB/T 1043	7.7 kJ/m²

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu P1S printer (nozzle temperature: 285 °C, bed temperature: 100 °C, printing speed: XY—155 mm/s、Z—147 mm/s) and IPCON ASA CF (Black) filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

1. Top UV resistance, and weather resistance, outstanding wate rand heat resistance, low density, high rigidity, ultra-low warpage (no need for positive chamber heating), excellent dimensional stability and surface quality, compatible with AMS or CFS, capable of ultra-high-speed printing (up to 300 mm/s).
2. Garden plant pots, camping tool boxes, corridor lamps, bird cages, surfboard parts, swimming pool pipe fittings, weather monitor covers, ship parts, aircraft models, ship models and other items that are used in outdoor and water environments or require lightweight and high rigidity.

• Recommended Printing Settings

Drying Settings Before Printing	70 - 90 °C, 4 - 8 h (Blast drying oven)
Printing Temperature and Humidity	≤ 80 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Compatible Support Material	Itself or support filament for ASA
Compatible Printer Type	Enclosed-frame (recommended), open-frame,
AMS or CFS	Compatible
Compatible Nozzle Material	Hardened steel, diamond, tungsten carbide, etc
Compatible Nozzle Size	0.6 mm (recommended) / 0.4 mm / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plate
Plate Surface Preparation	Apply liquid glue or PVP solid glue stick if the first layer bonding is not firm enough
Nozzle Temperature	265 - 290 °C
Bed Temperature	90 - 110 °C
Fan Speed	10% - 40%
Max Print Speed	300 mm/s

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